

ADA: Demographic Methods

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Week 4: Fertility

Overview

- ▶ Demographic transition
- ▶ Measures of fertility
- ▶ Measures of reproduction
- ▶ Fertility models

The demographic transition

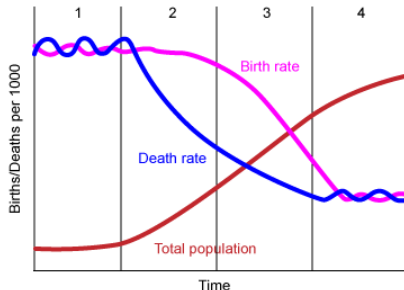
The demographic transition

Over time, human populations broadly go through four demographic stages:

1. High mortality / high fertility (Pre-transition)
 - ▶ high rate of infectious diseases, poor medical coverage
 - ▶ no modern contraception
 - ▶ agricultural economies, non-nuclear families
 - ▶ population size: low and stable
2. Falling mortality (Early transition)
 - ▶ technological progress (vaccinations, modern medicine)
 - ▶ improved infrastructure (sanitation)
 - ▶ population size: increasing rapidly
3. Falling fertility (Late transition)
 - ▶ empowerment of women
 - ▶ modern contraception
 - ▶ urbanization
 - ▶ population size: increasing but deceleration

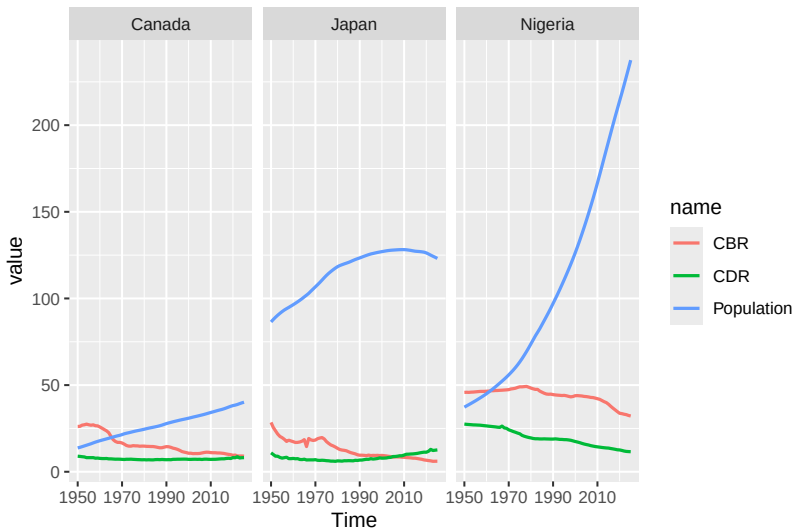
The demographic transition

4. Low mortality and fertility (Post-transition)
 - ▶ population size: high and stable
5. (TBD?) fertility < mortality
 - ▶ improvements in old-age mortality
 - ▶ children are expensive
 - ▶ population size: declining



Graph source: IUSSP

Examples



Measures of fertility

Note: fertility versus fecundity

- ▶ Fertility is studying the outcome (circumstances where live births occur)
- ▶ Fecundity is the studying the ability to conceive

Much of the earlier methodological work focused on fecundability (measuring the probabilities of conception), natural fertility (fertility in the absence of parity-specific control), waiting times to conception. . .

- ▶ treating conception as a random event
- ▶ ignoring conception out of marriage
- ▶ measuring base fecundity to determine the effectiveness of contraception

Crude measures

Crude birth rate:

$$CBR = \frac{\text{Births to women}}{\text{PY lived by population}}$$

But the population at risk: all women aged 15-49.

General fertility rate:

$$GFR = \frac{\text{Births to women}}{\text{PY lived by women of reproductive age}}$$

Recall discussion about why crude rates are good and bad.

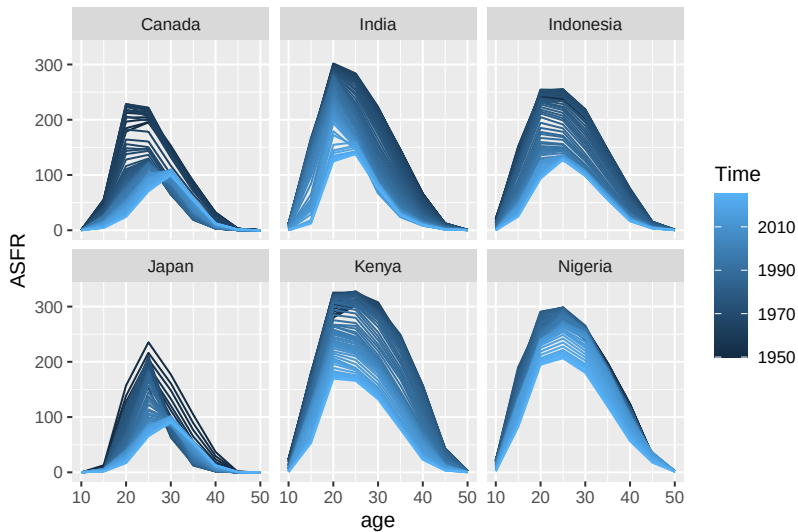
Age-specific fertility rates

$${}_nF_x = \frac{\text{Births to women aged } x \text{ to } x + n}{\text{PY lived by women aged } x \text{ to } x + n}$$

Continuous version:

$$f(x) = \frac{B(x)}{P(x)}$$

Age-specific fertility rates



Total fertility rate

TFR is the average number of babies a women would have if she lived through the entire reproductive lifespan.

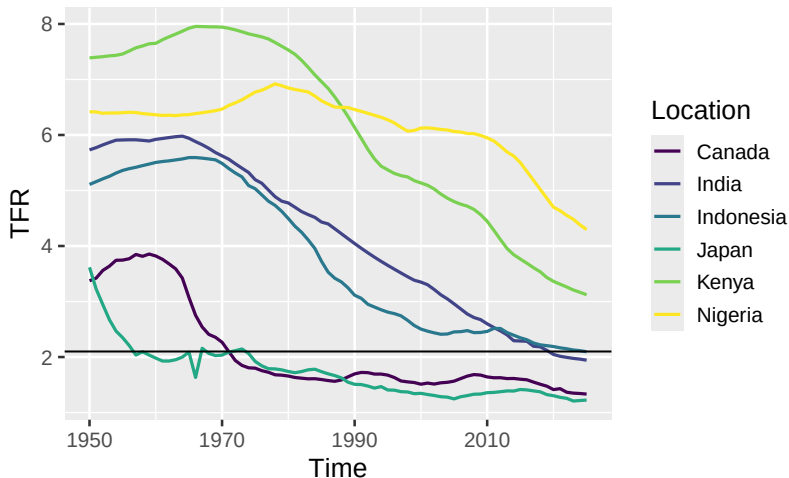
$$TFR = n \cdot \sum_{15}^{49} {}_nF_x$$

or

$$TFR = \int_{15}^{49} f(a) da$$

Example

Total fertility rate for selected countries, 1950–2015



Replacement level fertility

- ▶ The level of fertility required for a generation to replace itself
- ▶ $TFR \approx 2.1$, but varies by region
- ▶ If $TFR < 2.1$ does not mean population is declining

Parity

- ▶ Sometimes we get data in the form of 'children ever born' to a women (common in surveys)
- ▶ **Parity** is the number of live births a woman has had.
- ▶ A woman is **nulliparous** if they have parity = 0.
- ▶ Studying fertility by parity gives us an idea of how women are limiting their fertility
- ▶ Cohort measure

Define $w(j)$ to be the number of women at parity j

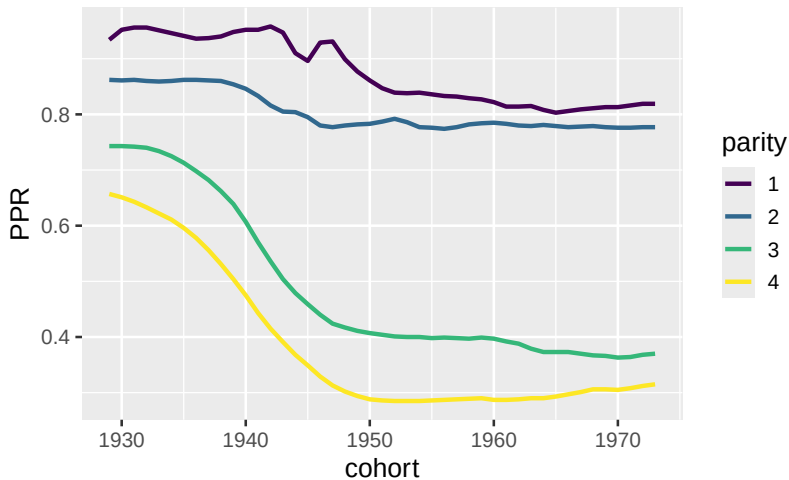
Parity progression ratios

Parity Progression Ratio at parity j $PPR(j)$ is the fraction of women who, having reached parity j , go on to have at least another child.

$$PPR(j) = \frac{\sum_{j+1}^{\infty} w(i)}{\sum_j^{\infty} w(i)}$$

Example

Parity Progression Ratios in Canada



Measures of reproduction

Measures of reproduction

The idea of generational renewal: compare sizes of successive cohorts of women

Only want to consider female babies.

Define fraction female at birth $f_{fab} \approx 0.4886$ i.e. there are slightly fewer girls born c.f. boys.

Gross reproduction ratio

GRR is the average number of **female** babies a women would have if she lived through the entire reproductive lifespan.

$$GRR = n \cdot \sum_{15}^{49} {}_nF_x^F$$

Or, if we don't know sex-specific fertility rates

$$GRR = n \cdot \sum_{15}^{49} {}_nF_x \cdot f_{fab}$$

Net reproduction ratio

NRR is the average number of female babies a women would bear if they were subject to the observed age-specific mortality rates.

$$NRR = \sum_{15}^{49} {}_nF_x^F \cdot {}_nL_x$$

or

$$NRR = \sum_{15}^{49} {}_nF_x \cdot {}_nL_x \cdot f_{fab}$$

Note the ${}_nL_x$ refers to the survivorship of women.

In continuous form, the product of fertility rates (of female babies) and the survivorship is given its own notation, $\phi(x)$, and called the net maternity function. Then

$$NRR = \int \phi(a) da$$

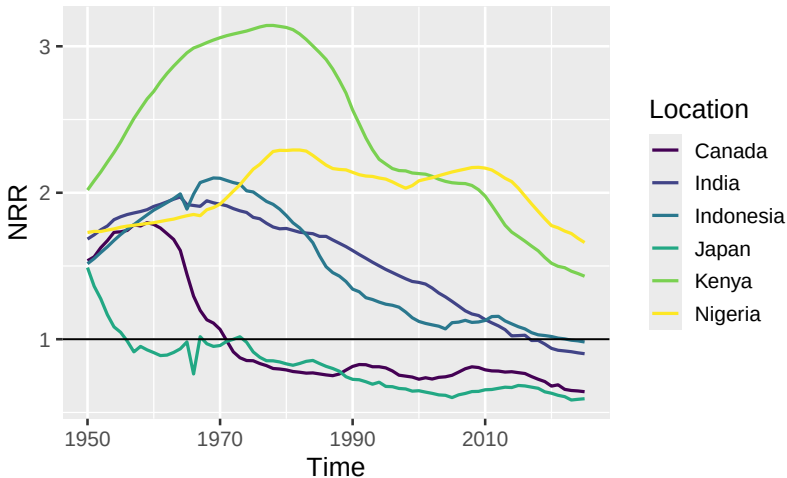
NRR from parity

If we are given data in the form of number of women by parity in a cohort, then we can also calculate the NRR as:

$$NRR = \frac{\sum_0^{\infty} i \cdot w(i)}{\sum_0^{\infty} w(i)}$$

Examples

NRRs for selected countries



How does NRR relate to population growth?

- ▶ $\text{NRR} > 1$ implies?
- ▶ $\text{NRR} < 1$?
- ▶ $\text{NRR} = 1$?
- ▶ How does this relate to replacement level fertility?

More next week (population growth, stable populations)

Mean age at child-bearing

$$\mu = \frac{\sum_x x \cdot {}_nF_x}{\sum_x {}_nF_x}$$

or

$$\mu = \frac{\int_x af(a)da}{\int_x f(a)da}$$

Fertility models

Fertility distributions are different to mortality

- ▶ repeated events
- ▶ separated by intervals of non-risk
- ▶ individual control

Harder (and less appropriate?) to model at the aggregate level than mortality.

Model fertility schedules

Coale and Trussel (1974). Focus:

- ▶ capture the extent to which observed mortality deviates from natural fertility
- ▶ use age-specific standards that are empirically derived (similar idea to Brass mortality)
- ▶ Parameterize in terms of overall level of fertility and degree of fertility control
- ▶ Note: estimating marital fertility. (!)

Model fertility schedules

$${}_nF_x = M \cdot n(x) \cdot e^{-m \cdot \nu(x)}$$

- ▶ M is the overall level of fertility
- ▶ $n(x)$ is the natural fertility age-specific schedule (constant)
- ▶ m is the strength of parity-specific limitation
- ▶ $\nu(x)$ is the impact of fertility limitation by age (constant)

Natural fertility schedule $n(x)$

Derived from the Hutterites (anabaptists, community of goods, limited use of technology). High fertility rates, closest thing to natural fertility observed (marry young, religious duty)

Interestingly, now evidence Hutterite fertility is declining.



Natural fertility schedule $n(x)$

Age	$n(x)$
15	0.36
20	0.46
25	0.431
30	0.396
35	0.321
40	0.167
45	0.024

What's the implied TFR?

Fertility limitation schedule $\nu(x)$

Derived from empirical data

Age	$\nu(x)$
15	0
20	0
25	0.279
30	0.667
35	1.042
40	1.414
45	1.67

Estimating M and m

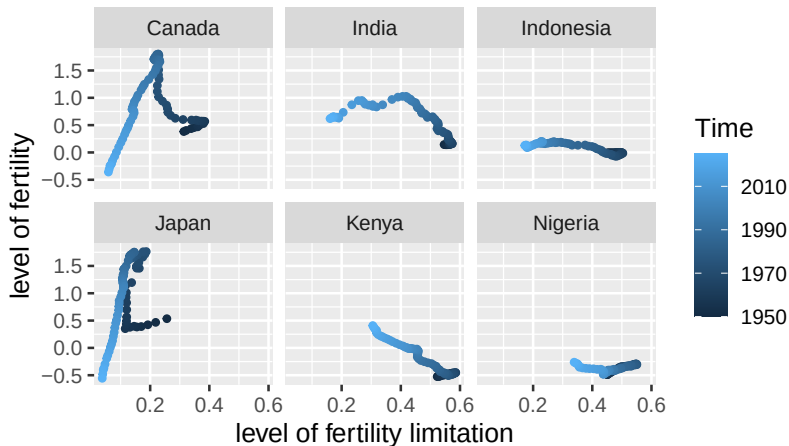
$${}_nF_x = M \cdot n(x) \cdot e^{-m \cdot \nu(x)}$$

implies

$$\log \left(\frac{{}_nF_x}{n(x)} \right) = \log M - m \cdot \nu(x)$$

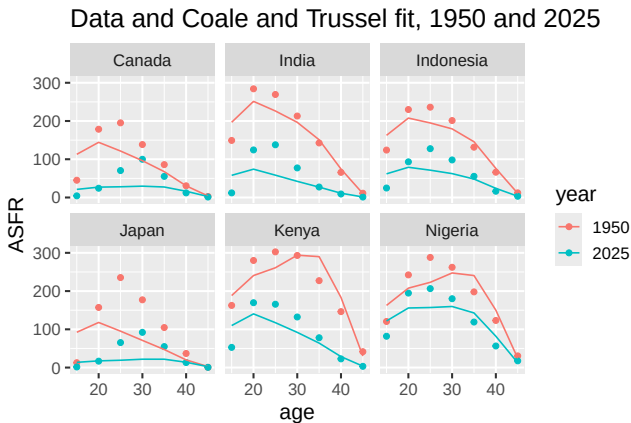
Example

Coale and Trussell's M and m for selected countries
1950–2025



Example

BUT the fits are pretty bad



..ideas for how to do this differently?

Some alternatives

- ▶ Schmertmann (2003) quadratic splines model
- ▶ Parametric distributions (skewed Normal and extensions):
e.g. see Mazzuco and Scarpa (2015)
- ▶ SVD!!!!

Tempo and quantum

Background

Note: shift in focus of research from high to low fertility.

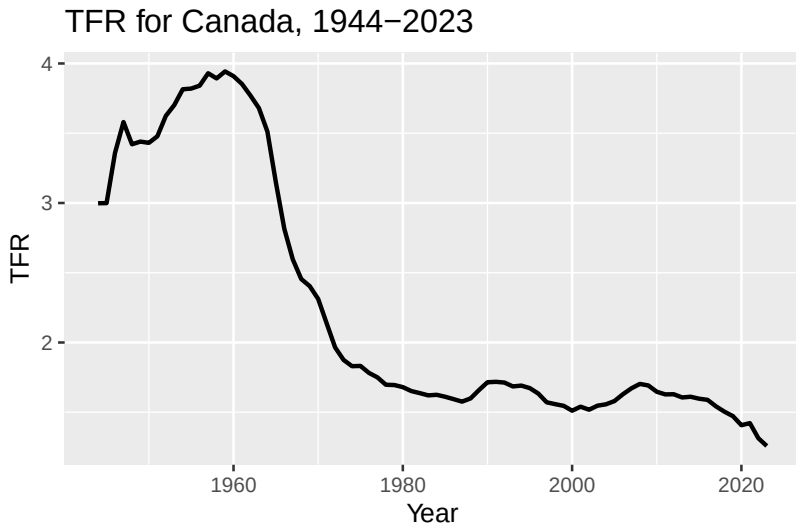
Most common measure of fertility is the period TFR. Is TFR declining because women are having fewer kids or because they are putting it off until later?

In any given year, fertility rates can decline because of two reasons:

- ▶ Quantum: decline in number of children
- ▶ Tempo: shift in age at childbearing (delaying having kids to older ages)

Note: this isn't an issue with cohort TFR, just period TFR

Background



Tempo-adjusted TFR

$$TFR'(t) = \frac{TFR(t)}{1 - \frac{d}{dt}\mu(t)}$$

$\mu(t)$ is the mean age at child-bearing in period t .

How to calculate this in practice?

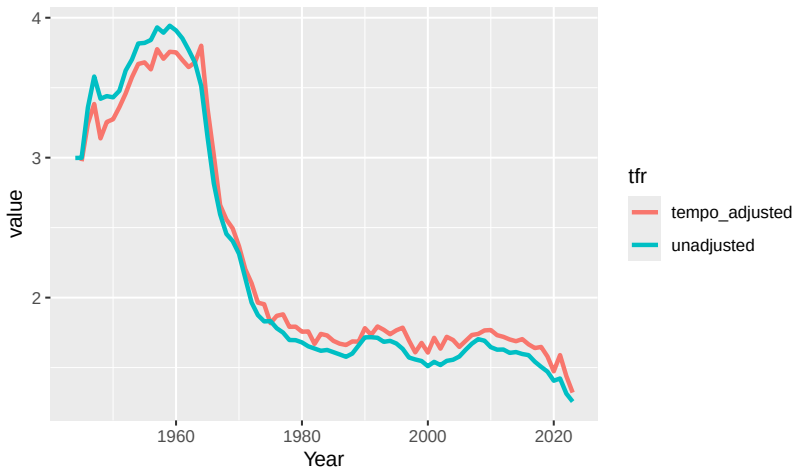
Bongaarts and Feeney (1998) calculate the change in the mean age at child-bearing by parity i , r_i :

$$TFR'(t)_i = \frac{TFR_i}{1 - r_i}$$

then $TFR'(t) = \sum_i TFR'_i$

Example

TFR and tempo-adjusted TFR
Canada 1944–2023



Fertility policies and interaction with societal forces

Trump Wants Americans To Make More Babies. Critics Say His Policies Won't Help Raise Them.

By [Stephanie Armour](#) and [Amanda Seitz](#)

DECEMBER 3, 2025



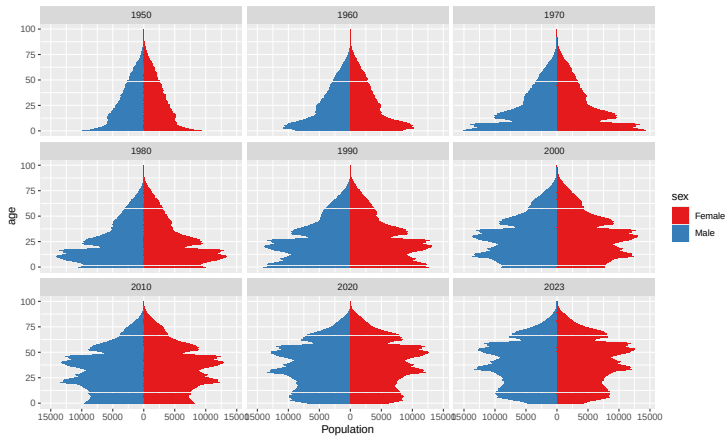
REPUBLISH THIS STORY



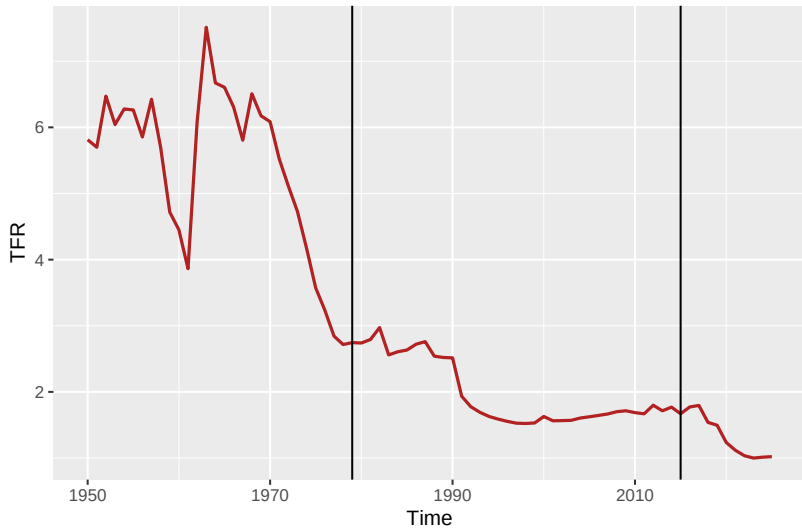
DISPONIBLE EN ESPAÑOL

On the other end of the scale (China's One Child Policy, 1979-2015)





Total fertility rate



Just this weekend (in FT)

Chinese baby bust reflects ageing society and reluctance to marry

Fertility trap?

European Journal of Population (2025) 41:32
<https://doi.org/10.1007/s10680-025-09755-5>

ORIGINAL RESEARCH

Only Children and Low Family Size Ideals: Did the One-Child Policy Create a “Low-Fertility Trap” in China?

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Abstract

The factors that shape fertility preferences—and their transition to reality—have been widely discussed. However, very few empirical studies have estimated the causal effect of sibship size on fertility preferences. Using the case of urban China, this study examines if growing up as an only child can lead to lower fertility ideals. Exploiting the introduction of the one-child policy in 1980 and using a fuzzy regression discontinuity design, the study finds that, among individuals born right around 1980, the increased probability of being an only child significantly reduces the ideal number of children and the probability of desiring two or more children. The sibship size effect not only offers a plausible explanation for low fertility ideals in urban China but also attests to a key mechanism underlying the “Low-Fertility Trap” hypothesis.

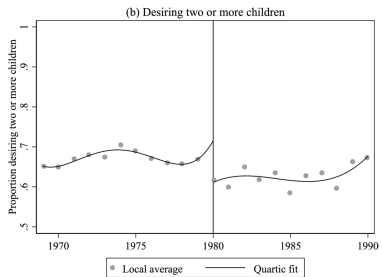
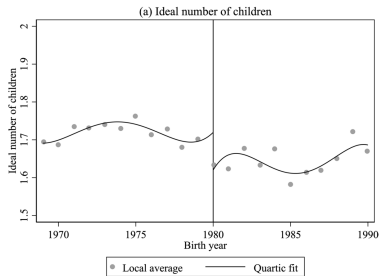


Table 3 Reduced form: Estimated discontinuity in personal ideal family size (also known as intent-to-treat effect of the OCP on personal ideal family size)

	(1)	(2)	(3)
	Discontinuity in ideal family size		
$\hat{\beta}_2$	-0.099** (0.035)	-0.080** (0.024)	-0.058* (0.025)
	Discontinuity in desiring two or more children		
$\hat{\beta}_2$	-0.104*** (0.016)	-0.077*** (0.007)	-0.060*** (0.006)
Birth year polynomial	Quartic	Quadratic	Quadratic
Window	[1969, 1990]	[1975, 1984]	[1975, 1984]
Covariates	No	No	Yes
N	8,696	3,952	3,952

Standard errors are in parentheses and clustered by birth year. All models control for birth year polynomial that is allowed to be different before and after 1980. Covariates included in Model 3 are whether mother has high school education or above, whether mother is a member of the Chinese Communist Party, whether the individual has non-agricultural hukou at birth, age of the individual at the time of the survey, fine rates on excess births in 1979 (Ebenstein, 2010), and region fixed effects. *** $p < 0.001$

** $p < 0.01$ * $p < 0.05$